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Brief Paper

State-space representation for fractional order controllers☆

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Abstract

This article proposes an infinite-dimensional state-space realization for linear filters with transfer function $C_d(s) \triangleq C_0((1 + s/\omega_b)/(1 + s/\omega_h))^d$, where $0 < \omega_b < \omega_h$ and $0 < d < 1$. This exponentially stable representation is derived from the Taylor expansion at zero of the function $(1 - z)^d$, and is made up of an infinite number of first-order ordinary differential equations. Finite-dimensional approximations obtained by truncating this representation are shown to converge towards C_d in H_∞. An example of feedback loop incorporating this approximation of C_d (car suspension) is presented, for which robustness of closed-loop resonance and step response overshoot vis-à-vis a variation in the vehicle mass is achieved. © 2000 Elsevier Science Ltd. All rights reserved.

Keywords: Distributed-parameter systems; Linear systems; Robust control; State-space realization

1. Fractional filters as tools for controller design

This article introduces a simple state-space representation and an approximation technique for linear filters with a transfer function of the form

$$C_d(s) \triangleq C_0\left(\frac{1 + s/\omega_b}{1 + s/\omega_h}\right)^d, \tag{1}$$

where $0 < \omega_b < \omega_h$, $C_0 > 0$ and $0 < d < 1$. In order to motivate the use of those so-called fractional filters in control applications, which has been pioneered by Oustaloup (see, e.g., Oustaloup, Mathieu & Lanusse, 1995; Oustaloup, Moreau & Nouillant, 1996), consider the feedback loop in Fig. 1. A robust control problem of interest is to find C guaranteeing robust Q-factor (amplitude magnification factor at resonance frequency) for the transfer functions from y_r to y and from v to y for all plants in the form $P = \alpha P_0$, with $\alpha_m \leq \alpha \leq \alpha_M$.

An ideal solution to this problem is to take as the nominal loop transfer function $L_0(s) \triangleq CP_0(s) = (s/\omega_0)^{-n}$, where $1 < n < 2$. Because the Nyquist plots of L and L_0 are similar, the phase margin and Q-factors for the closed-loop transfer functions from y_r to y and from v to y are then independent of α and uniquely determined by n. Furthermore, since the transfer function from y_r to y can be approximated by a second-order stable system, the maximum closed-loop step response overshoot is also quasi-independent of α. Thus, a change of α will result in a slower or faster but equally damped response, which is a desirable robustness property in some applications, such as the car suspension design problem presented in Section 3. To the authors' knowledge, this argument was first formulated by Bode, who dubbed this choice of L the ideal loop transfer function (Bode, 1945; see also Bergeon & Irving, 1990; Åström, 1997).

This ideal loop transfer function can be approximated by making $L_0(j\omega)$ close to $(j\omega/\omega_0)^{-n}$ around the cross-over frequency ω_0 for which $|L_0(j\omega_0)| = 1$. When $P(s) = Ms^{-m}$, this requires that $\arg[C(j\omega)]$ should be close to $m - 2 + \varphi/\pi$, where φ is the desired phase margin, in a given frequency range around ω_0. One could obviously seek to achieve this by using the rational lead-lag filter corresponding to (1) with $d = 1$, because $\arg[C_1(j\omega)]$ is close to $\arg[C_1(j\omega_0)] = \arcsin[(\omega_h/\omega_b - 1)/(\omega_h/\omega_b - 1)]$ around the frequency $\omega_0 = (\omega_b\omega_h)^{1/2}$. However, it is easily seen that for this choice of d the width of the frequency range over which the condition $\arg[C_1(j\omega)] \simeq \arg[C_1(j\omega_0)] = m - 2 + \varphi/\pi$ holds is entirely determined by the choice of φ, and thus

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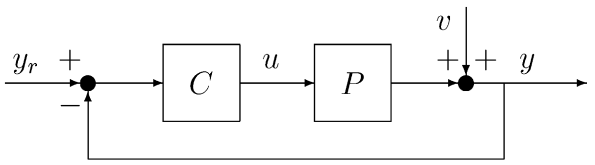


Fig. 1. Standard feedback loop with plant P, controller C, control input u, output y, reference y_r and disturbance v.

cannot be adjusted to meet robustness requirements. Conversely, allowing for non-integer values of d in (1) enables to guarantee robust phase margin, closed-loop resonance and overshoot for any given range of variation of α. To apply this control strategy to more general linear systems, one should take $C = C_d G$, where G has been chosen so that $GP_0(j\omega) \simeq M(j\omega)^{-m}$ around ω_0 (Bergeon & Irving, 1990).

Once an ideal C_d has been selected, it should be replaced by a rational approximation, to be used as the actual controller. The approach in Oustaloup et al. (1995, 1996) relies on a product of lead-lag filters whose zeroes and poles are recursively distributed in a frequency range around ω_0. These approximations are closely related to the so-called diffusive state-space representations (Mbodje & Montseny, 1995). The representation proposed here, which expands the results in Raynaud and Zergaïnoh (1997), is derived from a convergent Taylor expansion of the transfer function, and yields by simple truncation finite-dimensional approximations in a well-conditioned, minimal state-space form. The results presented here are easily extended to values of d outside $]0;1[$ and also to complex values of d.

Two qualifications should be added to these arguments. First, this approach focuses on a robustness criterion which may not be relevant in a number of applications. In fact controllers designed to meet it may prove very un-robust with respect to variations in the plant's dynamics, especially badly damped modes. Secondly, the problem of finding C so that $CP_0(j\omega) \simeq (j\omega/\omega_0)^{-n}$ could be solved by other methods than selecting and then approximating a fractional filter. However, on the condition that a simple approximation technique is available, this approach provides a "natural" and easy way to guarantee a robust closed-loop Q-factor.

2. State-space representation and approximation

The definition of C_d through (1) is ambiguous because $z^d \triangleq \exp(d \log z)$ is a multi-valued complex function. For C_d to be the transfer function of a linear filter, it should be the continuation of a real-valued function, so that C_d should be defined by using the principal value of z^d. In the sequel ℓ_2 denotes the set of vectors $x = (x_n)_{n>0}$ with bounded quadratic norm $\|x\| \triangleq (\sum_{n>0} x_n^2)^{1/2}$ and $\mathscr{L}(\ell_2)$ the set bounded linear operators from ℓ_2 unto itself.

Proposition 1. *Let $\sum_d$ be the linear system with state $x \in \ell_2$ defined by*

$$\dot{x}(t) = Ax(t) + Bu(t),$$

$$y(t) = Cx(t) + Du(t),$$

where $A \in \mathscr{L}(\ell_2)$, $B \in \ell_2$, $C \in \ell_2^*$ and $D \in \mathbb{R}$ are given by

$$A \triangleq -\omega_h I + (\omega_h - \omega_b) S_d, \qquad B_n \triangleq \frac{(\omega_h - \omega_b)}{n},$$

$$Cx \triangleq -dC_0 \left(\frac{\omega_h}{\omega_b}\right)^d x_1, \qquad D \triangleq C_0 \left(\frac{\omega_h}{\omega_b}\right)^d$$

with

$$(S_d x)_n \triangleq \frac{n-d}{n} x_{n+1}.$$

$\sum_d$ is an exponentially stable system whose transfer function is equal to C_d at least in the open half-plane $\mathbb{C}^+_{-\omega_h} \triangleq \{s \mid \Re(s) > -\omega_h\}$.

Proof. Denote the principal value of $(1-z)^d$ as $f(z)$, and the kth coefficient of the Taylor expansion of f at $z = 0$ as h_k, i.e.

$$h_0 \triangleq 1, \quad h_k \triangleq \prod_{l=0}^{k-1} \left(\frac{l-d}{l+1}\right) \quad \text{for } k \geq 1. \tag{2}$$

Since $\sum_{k\geq 0} h_k z^k$ coincides with $f(z)$ provided that $|z| < 1$, it follows that identity

$$C_0 \left(\frac{1 + s/\omega_b}{1 + s/\omega_h}\right)^d = C_0 \left(\frac{\omega_h}{\omega_b}\right)^d \left(1 - \frac{\omega_h - \omega_b}{s + \omega_h}\right)^d$$

$$= C_0 \left(\frac{\omega_h}{\omega_b}\right)^d \sum_{k\geq 0} h_k \left(\frac{\omega_h - \omega_b}{s + \omega_h}\right)^k$$

holds for all s outside the disk with center $-\omega_h$ and radius $(\omega_h - \omega_b)$. In this domain,

$$(sI - A)^{-1} = [(s + \omega_h) I - (\omega_h - \omega_b) S_d]^{-1}$$

$$= \frac{1}{s + \omega_h} \sum_{k\geq 0} \left(\frac{\omega_h - \omega_b}{s + \omega_h}\right)^k S_d^k, \tag{3}$$

where $S_d^k \triangleq S_d \circ S_d^{k-1}$. To prove that this identity makes sense, note that for all $k > 0$,

$$\forall n > 0, \quad (S_d^k x)_n = \prod_{l=0}^{k-1} \frac{n + l - d}{n + l} x_{n+k}$$

$$= \frac{(n+k) h_{n+k}}{n h_n} x_{n+k}. \tag{4}$$

Since $\|S_d\| \leq \sup_{n>0} |1 - d/n| \leq 1$, the series in (3) is convergent in the uniform topology in $\mathscr{L}(\ell_2)$ for s in $\mathbb{C}^+_{-\omega_h}$. Also, for s in $\mathbb{C}_0^+$

$$\|(sI - A)^{-1}\| \leq \frac{1}{\omega_h} \sum_{k\geq 0} \left(1 - \frac{\omega_b}{\omega_h}\right)^k = \frac{1}{\omega_b}.$$

Thus, $(sI - A)^{-1}$ belongs to $H_\infty(\mathscr{L}(\ell_2))$. This is an equivalent condition for the exponential stability of $\sum_d$ (Theorem 5.1.5 in Curtain & Zwart, 1995). Finally, since $h_0 = 1$ and $h_1 = -d$, it follows from (3) and (4) that for all s in $\mathbb{C}^+_{-\omega_h}$ the transfer function of $\sum_d$ verifies

$$C(sI - A)^{-1}B + D = C_0\left(\frac{\omega_h}{\omega_b}\right)^d\left[1 - d\sum_{k\geq 0}\left(\frac{\omega_h - \omega_b}{s + \omega_h}\right)^{k+1}\frac{h_{k+1}}{h_1}\right] = C_d(s).$$

Proposition 2. *Let $\sum_{dN}$ be the linear system with state $x \in \mathbb{R}^N$ defined by*

$$\dot{x}(t) = A_N x(t) + B_N u(t),$$

$$y(t) = C_N x(t) + Du(t),$$

where B_N, C_N are the truncation/restriction of B, C in/to $\mathbb{R}^N$ and $A_N \triangleq -\omega_h I + (\omega_h - \omega_b)S_{dN}$, with

$$(S_{dN}x)_n \triangleq \frac{n-d}{n}x_{n+1} \quad \text{if } n < N \quad \text{and} \quad (S_{dN}x)_N \triangleq \varepsilon x_N.$$

For all $\varepsilon \in [0,1]$, $\sum_{dN}$ is an exponentially stable system whose transfer function C_{dN} verifies

$$\frac{\|C_d - C_{dN}\|_\infty}{\|C_d\|_\infty} \leq 2\left(\frac{\omega_h}{\omega_b}\right)^d |h_N|\left(1 - \frac{\omega_b}{\omega_h}\right)^{N+1} \underset{N\to+\infty}{\sim} \Lambda\left(1 - \frac{\omega_b}{\omega_h}\right)^N N^{-d} \tag{5}$$

for some bound Λ independent of N, where $\|\cdot\|_\infty$ is the H_∞ norm over $\mathbb{C}^+_0$. Furthermore, there exists a value of $\varepsilon \in [0,1]$ such that $C_{dN}(0) = C_d(0) = C_0$.

Proof. It is immediately checked that A_N is an upper triangular matrix with negative diagonal terms, so that $\sum_{dN}$ is exponentially stable. Eq. (4) is replaced by

$$(S^k_{dN}x)_n = (S^k_d x)_n \quad \text{if } n + k \leq N,$$

$$(S^k_{dN}x)_n = \frac{Nh_N}{nh_n}\varepsilon^{n-N+k}x_N \quad \text{if } n + k > N.$$

Replacing S_d by S_{dN} in (3), and using the results in the proof of Proposition 1, one gets for all s in $\mathbb{C}^+$

$$C_{dN}(s) - C_d(s) = C_0\left(\frac{\omega_h}{\omega_b}\right)^d h_N\left(\frac{\omega_h - \omega_b}{s + \omega_h}\right)^{N+1} \times \sum_{k>N}\left(\frac{\omega_h - \omega_b}{s + \omega_h}\right)^{k-N-1}\left(\varepsilon^{k-N} - \frac{h_k}{h_N}\right). \tag{6}$$

Noting that $0 < h_k/h_N \leq 1$, $\|C_d\|_\infty = C_0(\omega_h/\omega_b)^d$, and taking the supremum of this upper bound over $\mathbb{C}^+_0$ leads to the requested H_∞ error bound. The asymptotic equivalent in (5) stems from the result (Flajolet & Odlyzko, 1990)

$$h_N \underset{N\to+\infty}{\sim} \frac{N^{-(d+1)}}{\Gamma(-d)},$$

where Γ is Euler's gamma function. Finally, from (6) and the fact that $0 < h_k/h_N \leq 1$, the function $g(\varepsilon) \triangleq C_d(0) - C_{dN}(0)$ is a strictly increasing continuous function which verifies $g(0) \leq 0$ and $g(1) > 0$. Consequently, there exists a unique $\varepsilon \in [0,1]$ such that $C_{dN}(0) = C_d(0) = C_0$. This completes the proof. □

Another candidate approximation strategy would be to construct $\sum_{dN}$ for a large value of N, and then to truncate it using a balanced reduced order approximation. However, it turns out that the balanced realization algorithm fails on $\sum_{dN}$ even for moderately high values of N. The failure seems to result from the fact that $\sum_{dN}$ has weakly controllable modes.

Obviously, the error bound in Proposition 2 is very conservative, which leads to suspect that prohibitively high values of N may be needed to obtain a reasonably accurate approximation. As a point of comparison, the rational approximations of order N obtained with the alternative technique proposed in Oustaloup et al. (1995, 1996) yields approximations accurate enough for control applications as soon as N is greater than 4 or 5. To get a more precise idea of the practicality of the approach in this article, the relative approximation errors $\|C_d - C_{dN}\|_\infty / \|C_d\|_\infty$ were computed as a function of N for the fractional controller used for the design example in Section 3, which corresponds to $d = 0.65$; $\omega_b = 0.5$; $\omega_h = 200$, and compared with the relative errors for the approximation C_N obtained from Oustaloup's method. The results for the values of N ranging from $N = 1$ to 100 are displayed in Fig. 2, using semi-logarithmic scale. For a given value of N Oustaloup's technique obviously yields a much better approximation. However, since in typical applications where C_N was used as the actual controller, satisfying performance was obtained with $N = 4$ or 5, these results also suggest that similar performance could be obtained by using the state-space approximation C_{dN} guaranteeing a comparable level of approximation error, i.e. with order in the range $N = 10$–20. Such controller order is by no means prohibitive, especially when the state-space representation is stable and well-conditioned.

3. Design example: car suspension

The control problem in this section is adapted from Oustaloup et al. (1996). The system is a simple mechanical model for an element of car suspension. M is the mass supported by each wheel, which is assumed to be

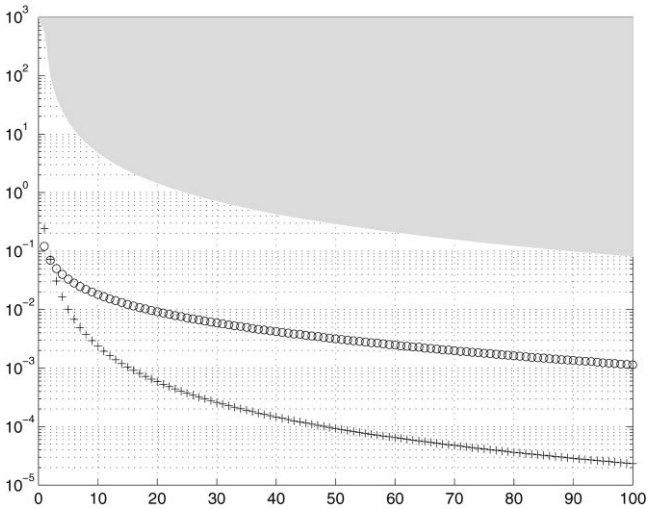


Fig. 2. Relative H_∞ approximation errors as a function of N for: o $= C_{dN}$; + $= C_N$; grey area = upper bound in formula (5).

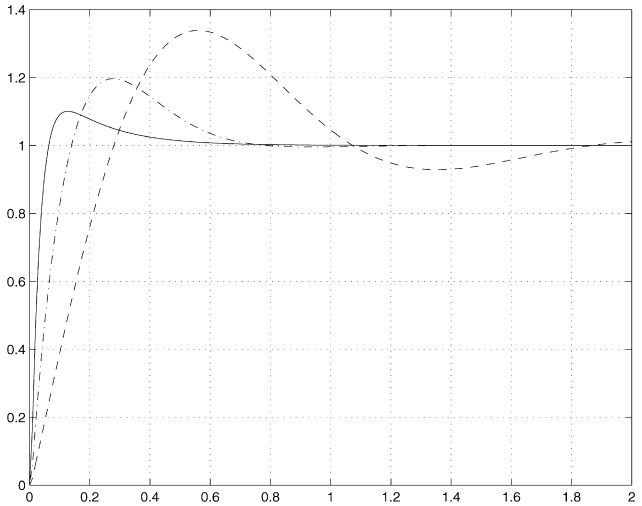


Fig. 3. Step responses of conventional suspension for $M = 100$, 300 kg (dash-dotted line) and $M = 900$ kg (dashed line).

equal to the fourth of total vehicle weight. The system input is the vertical position of wheel axis y_r, which is a function of road surface, vehicle speed and tire flattening. The system output y is the vertical positions of vehicle body. Both v and y are measured relatively to their respective equilibrium points. When a conventional suspension made up of a spring with stiffness k and a shock absorber with viscosity constant f is used, the Laplace transforms of v and y are related through equation

$$Ms^2Y(s) = -(k+fs)[Y(s) - Y_r(s)]$$

which corresponds to the feedback loop in Fig. 1, with $P(s) = 1/(Ms^2)$ and $C(s) = k + fs$.

Consider now the following robust control problem: design a suspension which ensures a relative step response overshoot of 20% for all values of M in the range $100\text{ kg} \leq M \leq 900\text{ kg}$, with constant closed-loop Q-factor, while the crossover frequency for $M = M_0 = 300$ kg is set to $\omega_0 = 10$ rad/s, in order to maximize passengers well-being. One immediately notices that satisfying both the crossover frequency and overshoot constraints for $M = M_0$ completely determines the value of the suspension parameters, which in this case is required to be approximately $k \simeq 1.74 \times 10^4$ and $f \simeq 3.5 \times 10^3$. Fig. 3 presents the closed-loop step response of the suspension for $M = 100$ kg; $M = 300$ kg; and $M = 900$ kg. Clearly, the overshoot is far from robust to the desired variation of M. Likewise, the closed-loop Q-factors vary from $Q \simeq 1.09$ for $M = 100$ kg to $Q \simeq 1.24$ for $M = 300$ kg and $Q \simeq 1.56$ for $M = 900$ kg.

The robustness requirements, however, are easily met by replacing the conventional suspension $C(s)$ by a fractional filter $C_d(s)$, and by setting $\omega_h = \lambda\omega_0$, $\omega_b = \omega_0/\lambda$, with $\lambda \gg 1$. When C_0 is chosen so that

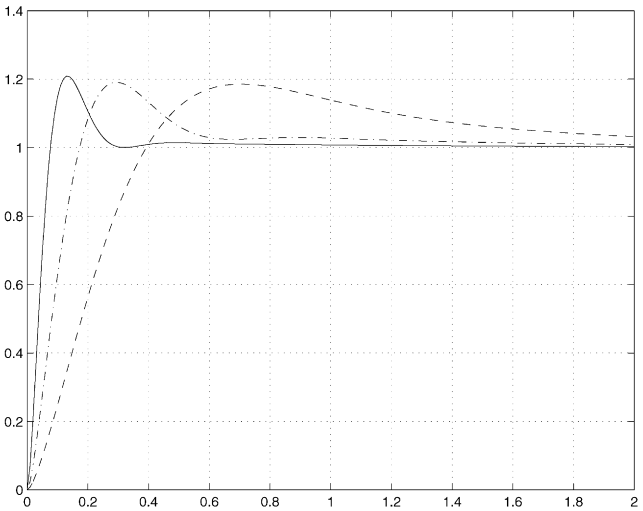


Fig. 4. Step responses of approximated fractional suspension for $M = 100$ kg, $M = 300$ kg (dash-dotted line) and $M = 900$ kg (dashed line).

$|P(j\omega_0)C_d(j\omega_0)| = 1$, i.e. $C_0 = M_0\omega_0^2\lambda^{-d}$, ω_0 becomes the gain crossover frequency for the nominal case $M = M_0$. Provided that $\lambda^{-1} \ll \omega_c/\omega_0 \ll \lambda$, the variation of M will result in a small variation of $\arg[PC_d(i\omega)]$, resulting in robust closed-loop Q-factor and overshoot. For the design example considered here, the value $\lambda = 20$ was selected, which corresponds to $\omega_b = 0.5$, $\omega_h = 200$. What remains to be done is to adjust, through closed-loop response simulations, the value of d in order to obtain a relative overshoot of 20%. This is achieved by setting $d = 0.65$. The corresponding closed-loop step responses, using the state-space approximation in Section 3 with $N = 20$, are presented in Fig. 4. The corresponding closed-loop Q-factors are near-constant, varying from

$Q \simeq 1.23$ for $M = 100$ kg to $Q \simeq 1.18$ for $M = 300$ kg and $Q \simeq 1.17$ for $M = 900$ kg.

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